

Lab Assignment 23.2 – AI Assisted Coding

M.Rushikesh

2303A51399

Lab Assignment 23.2 – Capstone Project Workshop

Task 1: Project Proposal

Question

Propose a real-world AI-assisted application.

Prompt

"Suggest a full-stack project idea with features, tools, and expected output."

Answer

Project Title: Student Attendance Tracker

Objective:

To manage and track student attendance digitally.

Features:

- Add student details
- Mark attendance
- View attendance reports
- Monthly summary

Tools/Technologies:

- Frontend: HTML, CSS, JavaScript
- Backend: Python Flask
- Database: SQLite

Expected Output:

- Fully functional attendance system
- Displays attendance reports

Code Explanation

- Defines project scope
 - Identifies tools and features
 - Forms base for development
-
-

Task 2: Application Design

Question

Design system architecture and database schema.

Prompt

"Generate system design and database schema for attendance tracker."

Code

```
CREATE TABLE Students (  
    id INTEGER PRIMARY KEY,  
    name TEXT  
);
```

```
CREATE TABLE Attendance (  
    id INTEGER PRIMARY KEY,  
    student_id INTEGER,  
    date TEXT,  
    status TEXT,  
    FOREIGN KEY(student_id) REFERENCES Students(id)  
);
```

Output

Tables created successfully.

Code Explanation

- Students table stores details

- Attendance table tracks records
 - Foreign key maintains relation
-
-

Task 3: AI-Assisted Code Development

Question

Develop frontend and backend.

Prompt

"Generate Flask backend and simple HTML frontend for attendance system."

Code

Backend (Flask)

```
from flask import Flask, request, jsonify
```

```
app = Flask(__name__)
```

```
attendance = []
```

```
@app.route('/mark', methods=['POST'])
```

```
def mark():
```

```
    data = request.json
```

```
    attendance.append(data)
```

```
    return jsonify({"message": "Marked"})
```

```
@app.route('/view', methods=['GET'])
```

```
def view():
```

```
    return jsonify(attendance)
```

```
app.run(debug=True)
```

Frontend (HTML)

```
<!DOCTYPE html>
```

```
<html>

<body>

<input id="name" placeholder="Name">

<button onclick="mark()">Mark</button>


<script>

function mark(){
  fetch('/mark', {
    method:'POST',
    headers:{'Content-Type':'application/json'},
    body: JSON.stringify({name: document.getElementById('name').value})
  });
}

</script>

</body>

</html>
```

Output

POST /mark →
{ "message": "Marked" }

Code Explanation

- Backend handles API requests
 - Frontend sends data using fetch
 - Integration via JSON
-

Task 4: Documentation & Code Quality

Question

Generate documentation and improve code readability.

Prompt

"Generate README and add comments to improve code quality."

Code

Attendance Tracker

Setup

1. Install Flask
2. Run app.py

Features

- Mark attendance
 - View records
-

Output

README created successfully

Code Explanation

- Provides setup instructions
 - Documents features
 - Improves maintainability
-

Task 5: Testing & Debugging

Question

Generate test cases and optimize code.

Prompt

"Generate unit test for attendance API."

Code

```
def test_mark():  
    data = {"name": "Alice"}  
    assert data["name"] == "Alice"
```

```
test_mark()  
print("Test Passed")
```

Output

Test Passed

Code Explanation

- Basic unit test
 - Validates input correctness
 - Helps debugging
-

Task 6: Deployment & Presentation

Question

Deploy application and present solution.

Prompt

"Explain deployment and presentation steps for project."

Answer

Deployment:

- Use GitHub for code hosting
- Deploy using GitHub Pages / Render / Heroku

Presentation Includes:

- Problem Statement
- Architecture Diagram
- Live Demo

- AI usage reflection
-

Code Explanation

- Deployment makes app accessible
 - Presentation explains project clearly
 - Shows end-to-end development
-